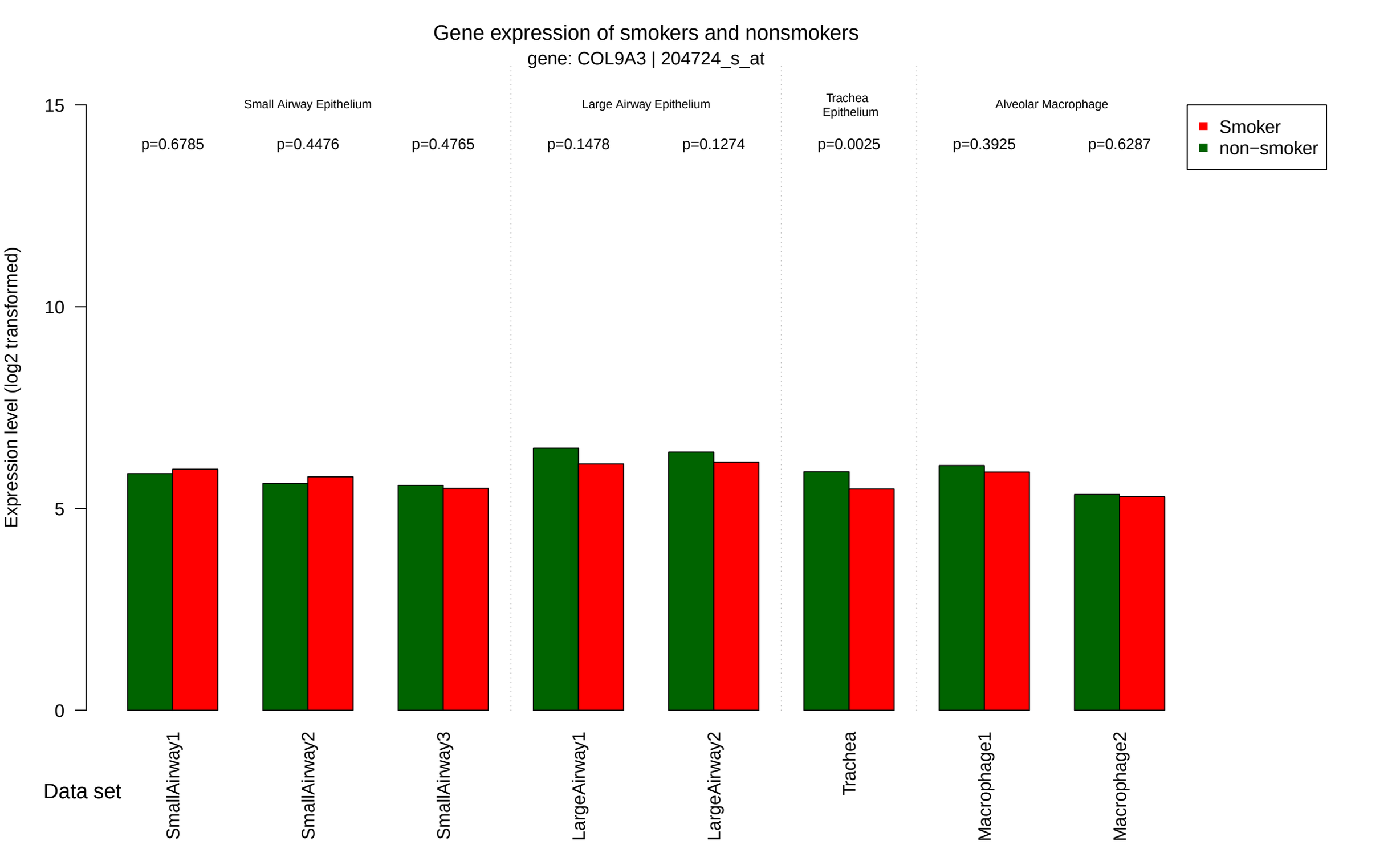
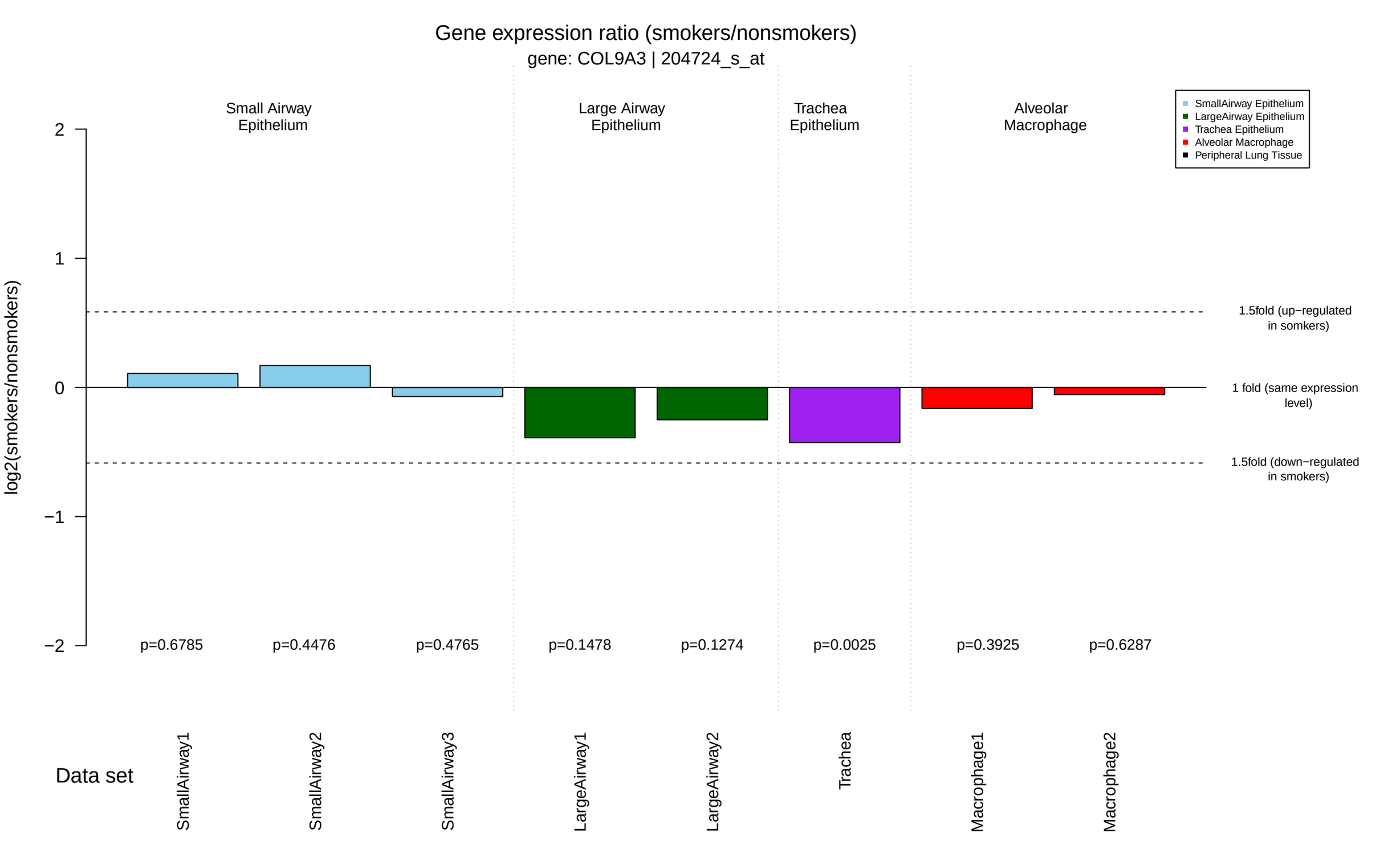
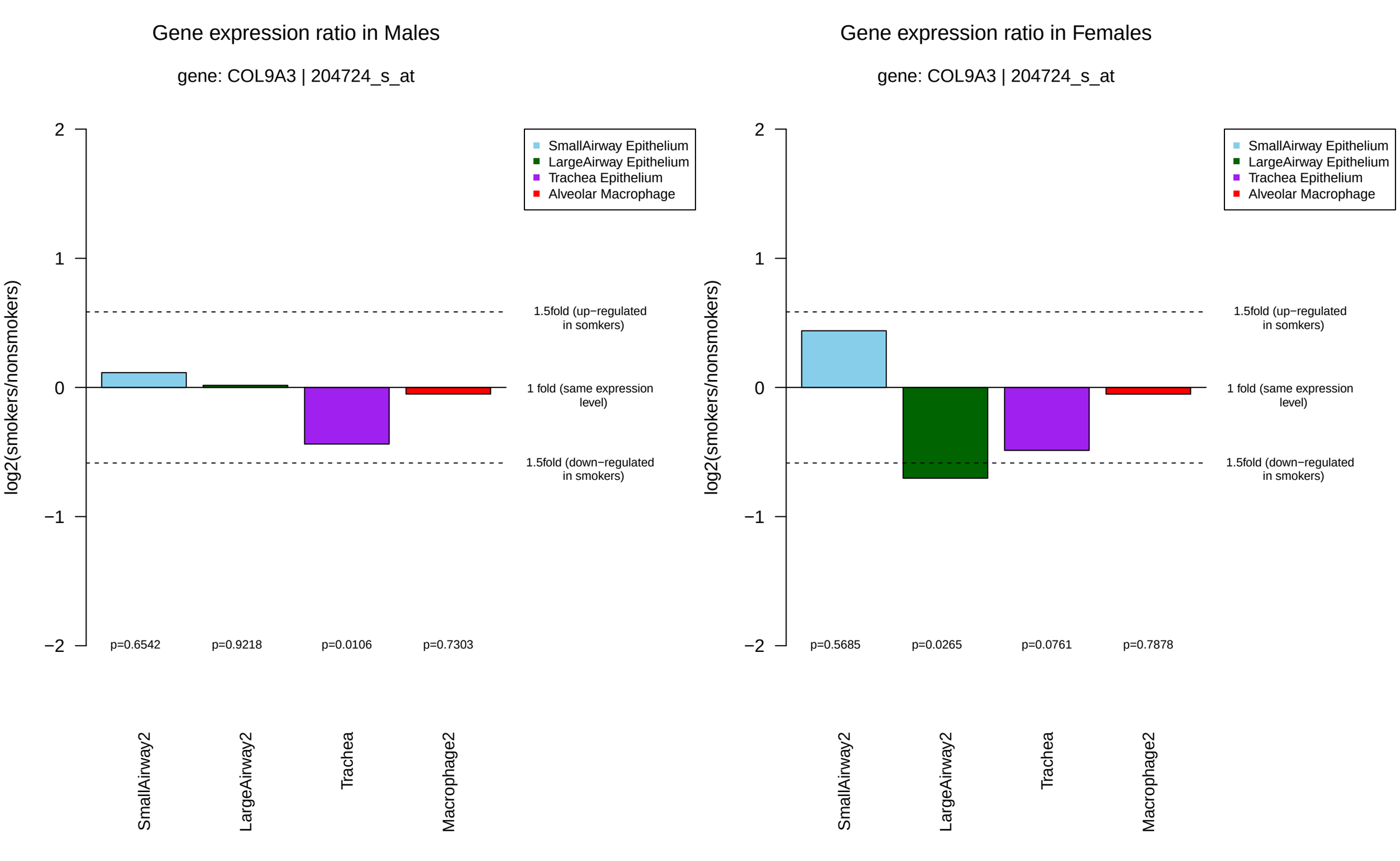
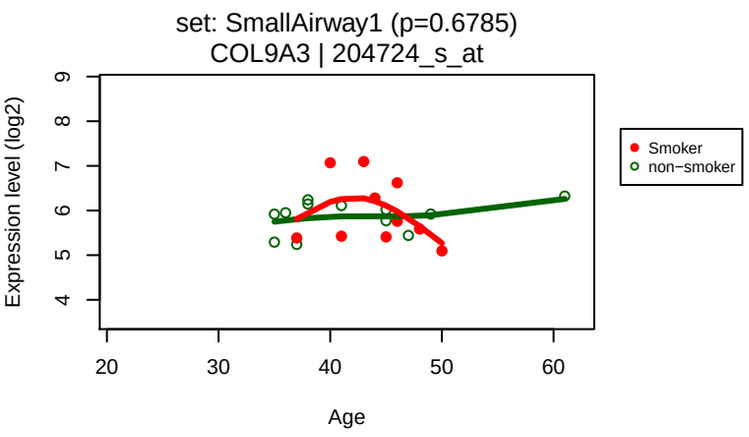
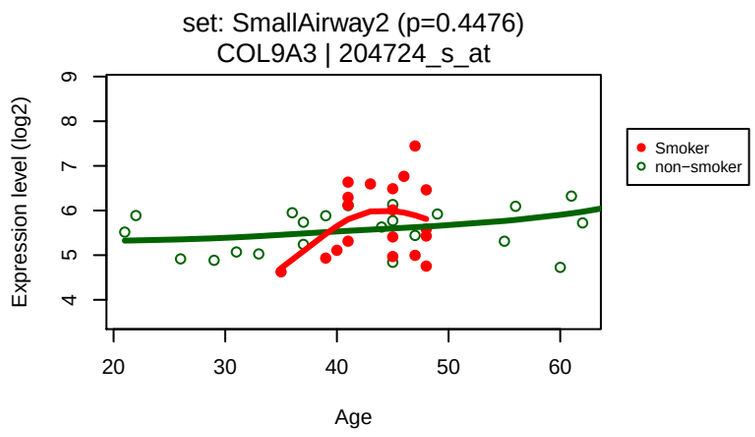
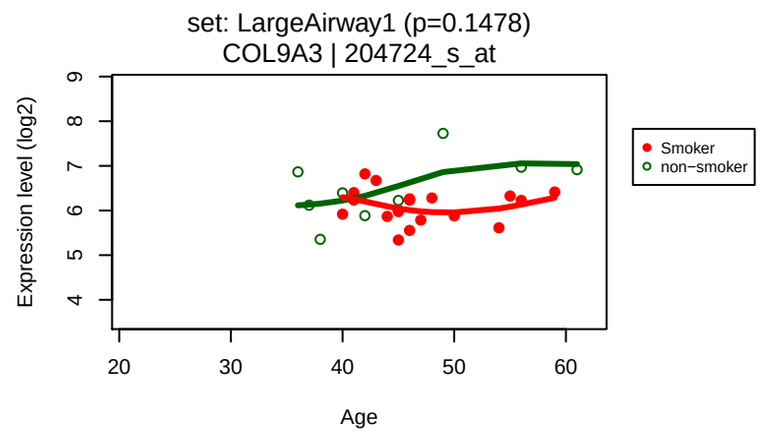

Gene expression of smokers and nonsmokers
gene: COL9A3 | 204724_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6785
p=0.4476
p=0.4765
p=0.1478
p=0.1274
p=0.0025
p=0.3925
p=0.6287
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

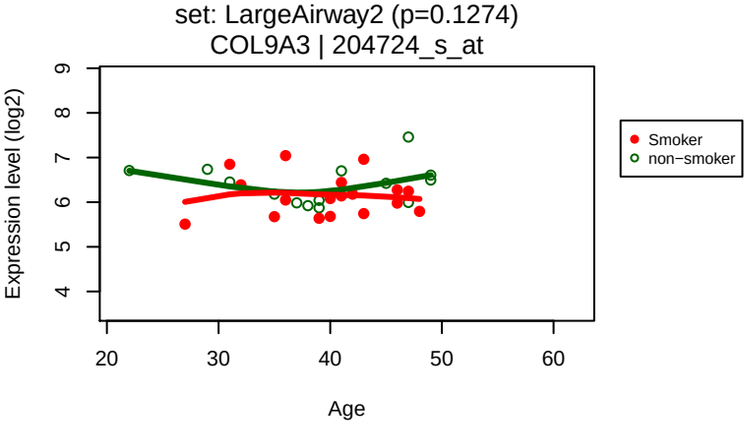

Gene expression ratio (smokers/nonsmokers)
gene: COL9A3 | 204724_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6785
p=0.4476
p=0.4765
p=0.1478
p=0.1274
p=0.0025
p=0.3925
p=0.6287
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

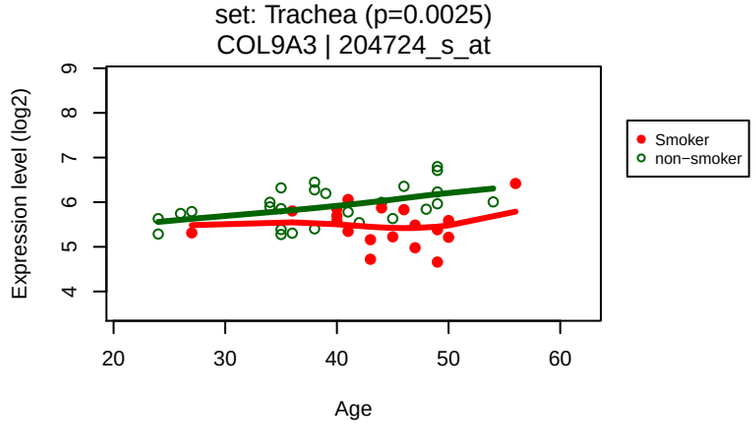

Gene expression ratio in Males
gene: COL9A3 | 204724_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6542
p=0.9218
p=0.0106
p=0.7303
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL9A3 | 204724_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5685
p=0.0265
p=0.0761
p=0.7878
SmallAirway2
LargeAirway2
Trachea
Macrophage2

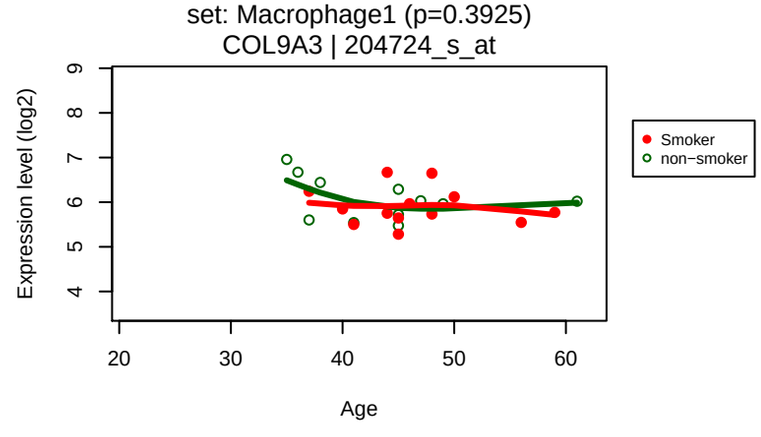

set: SmallAirway1 (p=0.6785)
COL9A3 | 204724_s_at
Expression level (log2)
Age
Smoker
non−smoker

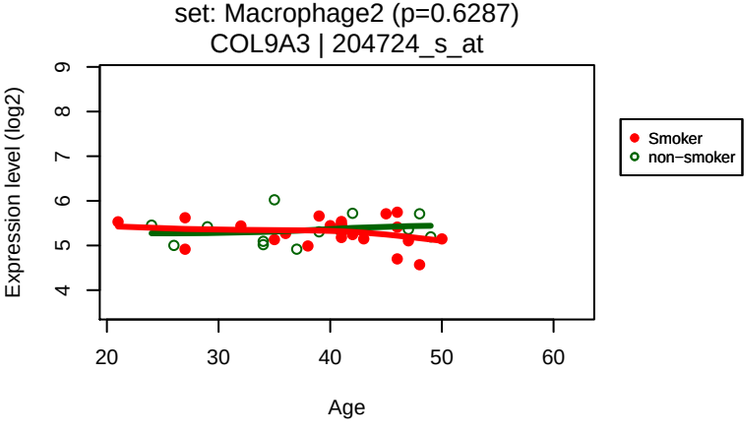

set: SmallAirway2 (p=0.4476)
COL9A3 | 204724_s_at
Expression level (log2)
Age
Smoker
non−smoker

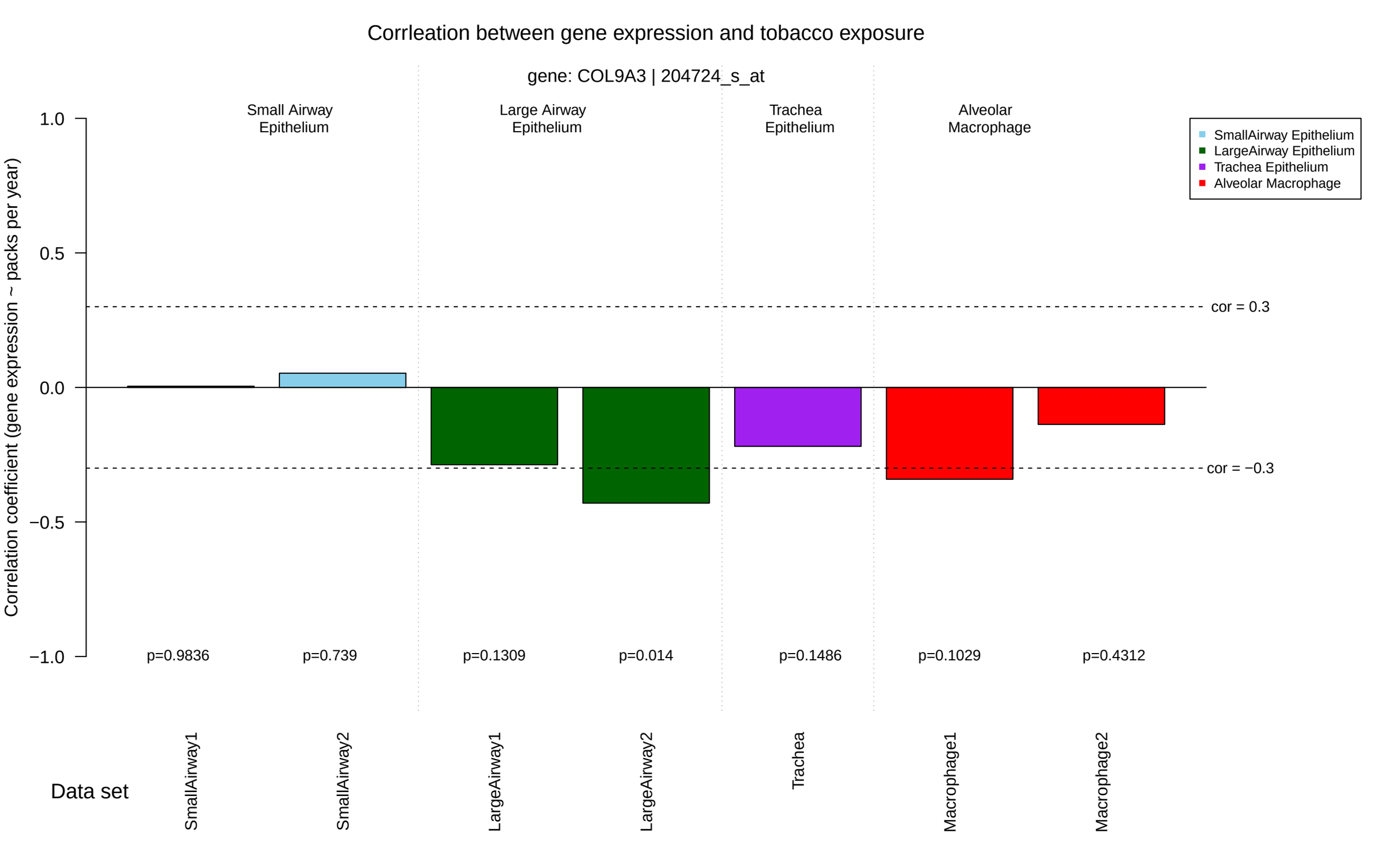

set: LargeAirway1 (p=0.1478)
COL9A3 | 204724_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1274)
COL9A3 | 204724_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0025)
COL9A3 | 204724_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3925)
COL9A3 | 204724_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6287)
COL9A3 | 204724_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL9A3 | 204724_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9836
p=0.739
p=0.1309
p=0.014
p=0.1486
p=0.1029
p=0.4312
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2